Valuation Sketchbook

Updated November 1, 2026

A beginner-first print companion: explanations, vocabulary, formulas, worked examples, checks, and the practice relevant to this subject.

How to use this edition

Read the everyday idea first. Define the words. Work through the example with your calculator. Attempt the check or question on paper before reading its explanation. Then change an input and explain what changes.

The web app adds narration, sliders, saved answers and timers. This edition includes paper experiments with predictions, calculations and explanations. It does not contain your personal browser notes.

Original study aid based on the supplied course materials. Final scope and assessment instructions may change. Risk uses reports/presentations in the supplied syllabus; Bloomberg certification is completed in its official portal.

Your reading path

1. Money today vs money later
2. Loans: payments, default and prepayment
3. Bonds and building a zero curve
4. Capital budgeting: NPV and IRR
5. Simulation, correlation and Cholesky
6. Random walks and stock price paths
7. Portfolio theory and the CAPM
8. ETFs and futures for end users
9. CDO waterfalls and tranches
10. Volatility smiles and implied densities
11. Real options and the Merton model

Then: paper labs where available, lesson-check answers, and subject practice.

Lesson 1: Money today vs money later

A dollar later is worth less than a dollar today, because today's dollar could sit in a bank and grow. Present value (PV) just undoes that growth.

Start with an everyday idea

You can choose $100 today or $105 next year. If your bank pays 5%, those choices have the same value in this simplified example.

Words you need

Present value means today’s equivalent amount. A rate of 5% is 0.05 in a formula. A cash flow is money paid or received.

Build the explanation

Think of the bank as a time machine for money. Put $V0 in at 5% and next year you get $V0 × 1.05. Present value runs the machine backwards: divide instead of multiply.

Future value (FV): what money grows into. Multiply by (1 + r) each year.

Present value (PV): what future money is worth today. Divide by (1 + r) each year.

Discount factor: the number you multiply a future cash flow by to get its PV. For 3 years at 5% it's $1/1.05^3 = 0.8638$.

Annuity: the same payment every period. Perpetuity: the same payment forever, PV = C / r.

Prof. Shimko's point ("PV fails"): PV only works cleanly when you know the cash flows and the right rate. Risky cash flows need a risk adjustment, which later classes cover.

Formula and units

PV = CF / (1 + r)^t

Perpetuity: PV = C / r

Annuity: PV = C × [1 - (1+r)^(-n)] / r

CF = cash flow, r = rate per period, t = number of periods, C = repeating payment.

Walk through the sketch

1. This is a timeline. Today is year 0. Someone promises you $100 at year 3.
2. Move the $100 back one year: divide by 1.05. It shrinks to $95.24.
3. Back one more year: divide by 1.05 again. Now $90.70.

4. Back to today: $86.38. That is the present value. Smaller coin, same "worth".

5. Shortcut: do all three divides at once. PV = 100 ÷ 1.05^3 = 86.38.

Worked example

You will get $100 in 3 years. The bank pays 5% a year. What is it worth today?

1. Write the formula: PV = 100 / 1.05^3

2. Work out 1.05^3 = 1.05 × 1.05 × 1.05 = 1.157625

3. Divide: 100 / 1.157625 = 86.38

4. Check by going forward: 86.38 × 1.157625 = 100. It works.

PV = $86.38

Common mistakes

1. Using an annual rate with monthly cash flows. Match the rate to the period (6% nominal annually with monthly compounding = 0.5% per month).

2. Mixing up multiply (future) and divide (present).

3. Forgetting that a perpetuity's first payment comes one period from now, not today.

Try before checking

What is $110 next year worth today at 10%?

1. $100

2. $110

3. $121

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_01_Lecture_1_Present_Value_Fails.pdf; FRE6103_BOOK_03_Valuation_Principles_Ch_1_2.pdf

Lesson 2: Loans: payments, default and prepayment

A loan payment is split into interest first, then the rest pays down what you owe. A pool of loans also loses money to people who default or pay off early.

Start with an everyday idea

When you repay a loan, part of each payment pays for borrowing and the rest reduces what you owe. Early payments often contain more interest because the outstanding balance is larger.

Words you need

Principal is the amount still owed. Interest is the borrowing charge. Default means failing to pay as agreed. Prepayment means paying earlier than scheduled.

Helpful earlier topics: Money today vs money later, How often interest is added

Build the explanation

This is the lecture's own example: borrow $5,000 for 24 months at 6% a year. Every month you pay the same amount. Early on most of it is interest, later most of it is principal, because the balance (and so the interest) keeps shrinking.

Amortization: paying the loan down to zero with equal payments.

Borrower's view: what does it cost me? Lender's view: what cash do I actually get back?

Default: some borrowers stop paying. In the course model, a fixed share of the remaining pool defaults each month (0.5% in Mini Project 1).

Prepayment: some borrowers pay everything early (1% a month in the project). The lender gets cash sooner but loses future interest.

Formula and units

Payment: PMT = B × i / [1 - (1+i)^(-n)]
Each month: interest = balance × i
principal paid = PMT - interest
new balance = balance - principal paid
Pool left after k months ≈ (1 - d - p)^k

B = amount borrowed, i = monthly rate (6%/12 = 0.5%), n = number of months, d = monthly default share, p = monthly prepay share.

Walk through the sketch

1. You borrow $5,000. That tall bar is what you owe at month 0.
2. Each month you pay $221.60. Red part = interest, blue part = paying down the loan.
3. Over 24 months the balance falls to zero.
4. Notice the red part shrinks and the blue part grows: less balance means less interest.
5. For a lender with a pool of loans, defaults and prepayments make the pool (dashed) shrink faster.

Worked example

Borrow $5,000 for 24 months at 6% a year. What is the payment, and what happens in month 1?

1. Monthly rate i = 6% / 12 = 0.005
2. PMT = 5000 × 0.005 / (1 - 1.005^(-24)) = $221.60 (Excel: PMT(0.06/12,24,-5000))
3. Month 1 interest = 5000 × 0.005 = $25.00
4. Principal paid = 221.60 - 25.00 = $196.60, new balance = $4,803.40

Payment $221.60; balance after month 1 is $4,803.40

Common mistakes

1. Using 6% instead of 0.5% for the month.
2. Forgetting interest is paid before principal.
3. In the pool model, default and prepayment both shrink the balance, but only default loses money.

Try before checking

A $100 balance earns $5 interest and you pay $20. What remains?

1. $85
2. $80
3. $105

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_03_Lecture_2A_Loans_with_exponential_default_risk.pdf; FRE6103_REVIEW_02_Mini_Project_1_Two_Sides_of_a_Loan_Instructions.txt

Lesson 3: Bonds and building a zero curve

A zero-coupon bond pays one amount at one date, so its price tells you today's value of $1 at that date. Bootstrapping peels these discount factors out of coupon bonds one date at a time.

Start with an everyday idea

A bond is an IOU: lend money now, receive promised payments later. Price each payment in today’s money and add them up.

Words you need

A coupon is a scheduled interest payment. Face value is the amount repaid at maturity. A zero-coupon bond has just one payment at the end. A discount factor converts future money into today’s money.

Helpful earlier topics: Money today vs money later, How often interest is added

Build the explanation

A coupon bond is just a bundle of little zero-coupon bonds, one for each payment. If you already know the discount factor for the early dates, you can subtract those pieces from the bond price. What's left must be the value of the last payment, which gives you the next discount factor. Repeat. That is bootstrapping.

Discount factor Z(t): price today of $1 paid at time t.

Zero rate: the rate that matches Z(t). Continuous: r = -ln Z(t) / t.

Zero curve / term structure: zero rates plotted against maturity. Mini Project 2 builds this from Treasury data.

Formula and units

Bond price = Σ CF(t) × Z(t)

Z(t) = e^(-r·t) → r = -ln Z(t) / t

Bootstrap last date: Z(T) = (Price - Σ earlier CF × Z) / CF(T)

CF(t) = cash flow at time t, Z(t) = discount factor, r = zero rate (continuous).

Walk through the sketch

1. Start with the easy one. A 6-month zero costs $98 for $100. So $1 in 6 months is worth $0.98 today.

2. Next, a 1-year coupon bond costing $99. It pays $2 at 6 months and $102 at 1 year.

3. We already know how to value the $2: 2 × 0.98 = $1.96.

4. The rest of the price must be the $102: 99 - 1.96 = 97.04. So Z(1) = 97.04 / 102 = 0.9514.

5. Turn each Z into a rate and plot. That is your zero curve.

Worked example

A 6-month zero sells at $98 per $100. A 1-year bond with a 4% coupon paid twice a year ($2 every 6 months) sells at $99. Find Z(0.5) and Z(1).

1. Zero bond: Z(0.5) = 98 / 100 = 0.98

2. The 1-year bond pays $2 at 0.5 years and $102 at 1 year.

3. Value of the first coupon: 2 × 0.98 = 1.96

4. What is left: 99 - 1.96 = 97.04 = 102 × Z(1), so Z(1) = 0.9514

5. Zero rates: -ln(0.98)/0.5 = 4.04%, -ln(0.9514)/1 = 4.98%

Z(0.5) = 0.98, Z(1) = 0.9514; zero rates 4.04% and 4.98%

Common mistakes

1. Forgetting the final payment includes the coupon (102, not 100).

2. Using the annual coupon (4) instead of the half-year coupon (2).

3. Mixing continuous and annual zero rates.

Try before checking

A zero-coupon bond pays $110 in one year. At 10%, its price is...

1. $100

2. $110

3. $121

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_04_Lecture_2B_Bonds_and_rates.pdf; FRE6103_DATA_08_Treasury_data_090426.xlsx; FRE6073_LECTURE_05_Lecture_04_Interest_Rates.pptx

Lesson 4: Capital budgeting: NPV and IRR

A company should take a project if its NPV is positive: the present value of what it brings in is more than what it costs. IRR is the discount rate where NPV is exactly zero.

Start with an everyday idea

Would you pay $100 now for a project that gives back $110 next year? At a required return of 5%, that future $110 is worth about $104.76 today, so the project adds about $4.76.

Words you need

NPV is present value of benefits minus costs. IRR is a discount rate that makes NPV zero. A required return is the rate used to judge the investment.

Helpful earlier topics: Money today vs money later, Bonds and building a zero curve

Build the explanation

NPV: add up the PV of every cash flow, including the (negative) cost today.

Discount rate: the cost of capital, i.e. the return investors could get elsewhere at the same risk (CAPM from the Portfolio class helps here).

IRR: the rate that makes NPV = 0. If IRR > cost of capital, NPV is positive. Find it by trial and error or Excel's IRR().

When IRR misleads: projects of different size, cash flows that change sign more than once (can have two IRRs), or different timing. When in doubt, trust NPV.

Payback period ignores timing and anything after the cutoff. Use it only as a rough check.

Welch's book also covers capital structure: how mixing debt and equity changes the cost of capital (WACC).

Formula and units

NPV = Σ CF_t / (1 + r)^t (t = 0, 1, 2, ...)

IRR: the r where NPV = 0

WACC = E/V × rE + D/V × rD × (1 - tax)

CF_0 is usually negative (the investment). E, D = market values of equity and debt, V = E + D.

Walk through the sketch

1. Cash flows on a timeline: pay $100 now, get 40, 50, 30 later.

2. Shrink each future amount to today's money at 8%, then add. NPV = +3.72.

3. Now try every discount rate. NPV falls as the rate rises.

4. Where the curve crosses zero (about 10.1%) is the IRR. Our 8% is to the left, so NPV > 0.

Worked example

A project costs $100 today and pays $40, $50 and $30 at the end of years 1, 2 and 3. Cost of capital 8%. NPV? IRR?

1. PV of year 1: 40 / 1.08 = 37.04

2. PV of year 2: 50 / 1.08^2 = 42.87

3. PV of year 3: 30 / 1.08^3 = 23.81

4. NPV = 37.04 + 42.87 + 23.81 - 100 = 3.72 → take it

5. IRR (Excel IRR or trial): about 10.1%, above 8%. Same answer.

NPV = $3.72, IRR ≈ 10.1%

Common mistakes

1. Forgetting the year-0 cost.

2. Choosing between two projects by IRR instead of NPV.

3. Using a company-wide discount rate for a project with very different risk.

Try before checking

Cost $100 today; receive $110 next year; discount rate 10%. NPV?

1. $10

2. $0

3. $110

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf; FRE6103_LECTURE_13_VL10_Multiperiod_cash_flows.pdf

Lesson 5: Simulation, correlation and Cholesky

Simulation means making up thousands of possible futures with random numbers and averaging the results. Cholesky is the trick that makes those random numbers move together.

Start with an everyday idea

Instead of predicting one future, roll many imaginary dice to generate possible futures. Then summarize the outcomes. Related risks need related dice, not independent ones.

Words you need

Simulation generates possible outcomes from assumptions. Correlation describes co-movement. Cholesky is a matrix method for giving random draws a chosen correlation structure.

Helpful earlier topics: Volatility, diversification, VaR and ES

Build the explanation

Some problems are too messy for a formula (10 bonds defaulting at different times, a CDO waterfall). So you let the computer roll dice many times and look at the outcomes.

Uniform U (between 0 and 1) → normal Z with NORM.S.INV(U) in Excel.

Correlation: independent Z's don't move together. To get correlation ρ, mix them: $X2 = \rho \cdot Z1 + \sqrt{(1-\rho^2)} \cdot Z2$. For many variables, the Cholesky matrix does this mixing for all of them at once.

Default time: with a constant default rate λ, a random default time is $\tau = -\ln(U)/\lambda$. If τ is before maturity, that bond defaults in this run.

Sanity checks: the sample mean, standard deviation and correlation should match what you asked for. "Moment matching" forces them to match exactly.

Formula and units

Two variables: $X1 = Z1$

$X2 = \rho \cdot Z1 + \sqrt{(1 - \rho^2)} \cdot Z2$

Many variables: $X = L \cdot Z$ where $L \cdot L^T$ = correlation matrix

Default time: $\tau = -\ln(U) / \lambda$

Z = independent standard normals, L = Cholesky (lower triangular) matrix, λ = default intensity per year.

Walk through the sketch

1. 400 pairs of independent random numbers. A round cloud: knowing one tells you nothing about the other.

2. Apply Cholesky with ρ = 0.6. The cloud stretches into a tilted oval. Now they move together.

3. One dot: (Z1, Z2) = (1.2, -0.5) becomes (1.2, 0.32). It moved up because Z1 was high.

4. Default times: draw U, then τ = -ln(U)/λ. Small U means a long wait; big U means early default.

Worked example

ρ = 0.6, and you drew Z1 = 1.2, Z2 = -0.5. Find X1 and X2. Then: λ = 4% a year and U = 0.7. When does the bond default?

1. X1 = Z1 = 1.2

2. √(1 - 0.36) = 0.8

3. X2 = 0.6 × 1.2 + 0.8 × (-0.5) = 0.72 - 0.40 = 0.32

4. τ = -ln(0.7)/0.04 = 0.3567/0.04 = 8.92 years. For a 5-year bond: no default in this run.

X1 = 1.2, X2 = 0.32; default at 8.92 years

Common mistakes

1. Using a covariance matrix when the formula needs a correlation matrix (or the other way round).

2. Re-drawing random numbers every time Excel recalculates. Fix them (paste values) so results are comparable.

3. Forgetting to check that sample correlation is near the target.

Try before checking

Why correlate simulated stock returns?

1. To guarantee a profit

2. To represent how assets move together

3. To remove uncertainty

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_05_Lecture_4_Simulation.pdf; FRE6103_DATA_02_Lecture_4_Cholesky_five.xlsx; FRE6103_DATA_03_Lecture_4_Simulation_spreadsheets.xlsx

Lesson 6: Random walks and stock price paths

A stock price is modelled as a steady drift plus random shocks. Using percentage shocks (geometric Brownian motion) keeps prices positive and gives a lognormal shape.

Start with an everyday idea

A stock path is like taking many small steps with both a general tendency and surprises. A simulated path is one possible story, not a price forecast you can rely on.

Words you need

Drift is the model’s average direction. Volatility sets the size of surprises. A time step is the interval between simulated prices. Brownian motion is a mathematical model of random changes.

Helpful earlier topics: How often interest is added, Simulation, correlation and Cholesky

Build the explanation

Wiener process (Brownian motion) W: a random walk where each tiny step is normal with variance equal to the time step. Over time Δt the change is ε√Δt.

Arithmetic Brownian motion: dX = μ dt + σ dW. Can go negative. Fine for spreads or rates, not for stock prices.

Geometric Brownian motion (GBM): dS = μS dt + σS dW. Returns are normal, prices are lognormal.

Simulating one step exactly: Snext = S × exp((μ - σ²/2)Δt + σ√Δt·Z). The -σ²/2 is there because of Itô's lemma; without it your average would be too high.

Mean reversion (e.g. rates, commodities): the drift pulls the value back toward a long-run level.

Formula and units

GBM: $dS = \mu \cdot S \cdot dt + \sigma \cdot S \cdot dW$

One step: $S(t+\Delta t) = S(t) \times \exp[(\mu - \sigma^2/2) \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot Z]$

Expected: $E[S_T] = S0 \times e^{(\mu T)}$

μ = expected return (drift), σ = volatility, Z = standard normal draw.

Walk through the sketch

1. Start at $100. Time runs left to right over one year.
2. One simulated path: small random up and down steps with a slight upward drift.
3. More paths. Each one is a possible future.

4. The blue line is the average path, 100·e^(0.08t). The band shows the typical spread, which widens over time.

5. At year end the prices pile up in a lopsided (lognormal) shape: a long tail upward, never below zero.

Worked example

S = 100, μ = 8%, σ = 20%, Δt = 1 year, and you draw Z = 0.5. What is the simulated price?

1. Drift part: (0.08 - 0.02) × 1 = 0.06

2. Shock part: 0.20 × √1 × 0.5 = 0.10

3. Add: 0.16, then e^0.16 = 1.1735

4. S = 100 × 1.1735 = 117.35

S = $117.35

Common mistakes

1. Forgetting -σ²/2 in the exponent.

2. Using σ × Δt instead of σ × √Δt.

3. Mixing up the median (S0·e^((μ-σ²/2)T)) and the mean (S0·e^(μT)).

Try before checking

Why do two simulated paths differ with the same starting price?

1. They use different random shocks

2. One must contain a mistake

3. The starting price determines the whole future

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_06_Lecture_5_Stochastic_Processes.pdf; FRE6103_DATA_05_Lecture_5_Simulation_spreadsheets.xlsx; Hull 8e, Chapter 13 (Wiener processes and Itô)

Lesson 7: Portfolio theory and the CAPM

Investors only get paid for risk they cannot diversify away. Beta measures that risk, and the CAPM turns beta into a required return.

Start with an everyday idea

Owning an umbrella shop and an ice-cream shop can smooth income because different weather helps each one. Diversification depends on how investments move together.

Words you need

A portfolio is a collection of investments. Beta measures sensitivity to market returns. CAPM links required return to market risk under simplifying assumptions; it is not a guaranteed return.

Helpful earlier topics: Volatility, diversification, VaR and ES

Build the explanation

Efficient frontier: the best possible return for each level of risk, built by mixing stocks (Markowitz). Correlation below 1 bends it to the left.

Capital market line: mix the risk-free asset with the best risky portfolio (the "market" M). It is the line from rf that just touches the frontier. Its slope is the Sharpe ratio.

Beta: how much an asset moves when the market moves. $\beta = \mathrm{Cov}(r_i, r_M)/\mathrm{Var}(r_M)$.

CAPM: $E[r] = r_f + \beta \times (E[r_M] - r_f)$.

Lintner (certainty-equivalent) form: value a risky cash flow X by subtracting a risk charge, then discounting at the risk-free rate. This is how the course values CDO tranches.

Formula and units

Sharpe ratio: $(E[r] - r_f) / \sigma$

CAPM: $E[r_i] = r_f + \beta_i \times (E[r_M] - r_f)$

Lintner: $PV = (E[X] - \lambda \cdot \mathrm{Cov}(X, r_M)) / (1 + r_f)$

$\lambda = (E[r_M] - r_f) / \sigma_M^2$

λ = market price of risk, X = an uncertain cash flow in one period.

Walk through the sketch

1. Each dot is one stock: its risk (σ) and expected return.

2. Mixing them gives the efficient frontier: the best return for each level of risk.

3. Add the risk-free asset at 4%. Draw a line from it that just touches the frontier.

4. The touching point is the market portfolio M. Everyone should hold a mix of M and the risk-free asset.

5. For one asset, CAPM says: required return = 4% + β × 6%.

Worked example

rf = 4%, market return 10%, σ_M = 20%. (a) Required return for β = 1.2? (b) A cash flow has E[X] = $110 and Cov(X, r_M) = 2. Value it with Lintner.

1. (a) 4% + 1.2 × (10% - 4%) = 11.2%

2. (b) λ = 0.06 / 0.04 = 1.5

3. Risk charge = 1.5 × 2 = 3, certainty equivalent = 110 - 3 = 107

4. PV = 107 / 1.04 = 102.88

Required return 11.2%; PV = $102.88

Common mistakes

1. Using total risk σ instead of beta for required return.

2. Discounting the certainty equivalent at a risky rate. It already includes the risk charge, so use rf.

3. Forgetting that correlation, not volatility alone, drives diversification.

Try before checking

Risk-free rate 3%, market premium 5%, beta 2. CAPM required return?

1. 8%

2. 13%

3. 10%

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_07_Lecture_6_MPT_and_the_CAPM.pdf; FRE6103_DATA_06_MPT_Ten_Stock_Portfolio.xlsx; FRE6103_DATA_07_sharpe_and_lintner.xlsx

Lesson 8: ETFs and futures for end users

Futures and ETFs look simple, but an end user faces hidden costs: rolling futures in contango, daily-reset leverage in leveraged ETFs, margin calls and basis risk.

Start with an everyday idea

An ETF is a basket you can buy as a share. A futures contract is an agreement with future settlement. Both can give market exposure, but their cash needs and risks differ.

Words you need

ETF means exchange-traded fund. Tracking error is deviation from a benchmark. Futures leverage means a small collateral balance supports a larger exposure.

Helpful earlier topics: Futures, margin and daily settlement, Portfolio theory and the CAPM

Build the explanation

ETF: a fund that trades like a stock and tracks an index. Low cost, but commodity ETFs usually hold futures, not the commodity.

Roll: futures expire, so a fund sells the near contract and buys the next one every month.

Contango (futures above spot, curve upward): each roll buys higher and the fund slowly loses (negative roll yield). Backwardation (curve downward) gives positive roll yield.

Leveraged ETFs (2×, 3×) reset every day. In a choppy market they lose value even if the index ends flat (volatility decay).

End-user risks: margin/liquidity calls, basis risk, volume (volumetric) risk when you do not know how much you will need to hedge.

Formula and units

Leveraged ETF over 2 days: $(1 + L \cdot r1)(1 + L \cdot r2) \neq 1 + L \cdot (\text{total index return})$

Roll yield (approx): $(F_{near} - F_{next}) / F_{near}$ per roll

L = leverage (2 for a 2× ETF), r = daily index return.

Walk through the sketch

1. An index that goes +10% then -10%: it ends at 99.

2. A 2× daily ETF: +20% then -20%. It ends at 96, worse than "2 × -1%".

3. Contango: futures prices rise with maturity. Each month the fund sells cheap near futures and buys pricier ones.

4. That gap is lost every roll: a slow leak called negative roll yield.

Worked example

An index goes +10% then -10%. What happens to the index and to a 2× daily ETF?

1. Index: 1.10 × 0.90 = 0.99 → down 1%

2. 2× ETF: day 1 = +20%, day 2 = -20%

3. 1.20 × 0.80 = 0.96 → down 4%

4. The ETF lost 4%, not 2× the 1% (2%). That extra loss is volatility decay.

Index -1%, 2× ETF -4%

Common mistakes

1. Expecting a 2× ETF to give 2× the return over months.

2. Ignoring roll cost when comparing a commodity ETF to the spot price.

3. Forgetting that futures hedges can create big cash margin calls even when the hedge works.

Try before checking

Which statement about futures margin is correct?

1. It is the maximum possible loss

2. More cash may be needed after adverse moves

3. It removes all investment risk

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_08_Lecture_7_ETFs_and_Futures_MARKUP.pdf; Futures lecture recording (see "Watch online")

Lesson 9: CDO waterfalls and tranches

A CDO pools many bonds and pays the cash out in order: senior tranche first, then junior, then whatever is left goes to equity. Who gets hurt depends on how many bonds default.

Start with an everyday idea

Think of several cups stacked under a tap. Payments fill the senior cup first. If there is not enough water, the lowest-priority cup goes dry first.

Words you need

A tranche is a slice with a payment priority. Senior is paid first. Equity takes the first losses. A waterfall is the rule for allocating cash and losses.

Helpful earlier topics: Credit risk: default, loss and spreads

Build the explanation

Collateral: the pool of bonds (Mini Project 3: ten $10m bonds, 6% coupons).

Waterfall: each period, pay tranche A what it is owed, then B, then the residual. No carry-forwards in the project version.

Senior (A) is safest; it only loses if many bonds default. Equity/residual takes the first losses.

Correlation matters: with high correlation, bonds tend to default together. That makes "almost nothing" and "lots" of defaults both likelier. Senior tranches get riskier; equity can get safer.

Value each tranche by simulating cash flows (Class 11) and pricing them with Lintner (Class 13).

Formula and units

Pay A = min(cash, owed to A)
Pay B = min(cash - Pay A, owed to B)
Residual = cash - Pay A - Pay B

Repeat every period in every simulation run, then average and discount.

Walk through the sketch

1. Three buckets stacked like a fountain: A on top, B in the middle, residual at the bottom. Cash pours in from the pool.

2. Good period: the pool pays $6m. A fills ($3m), B fills ($2m), $1m spills into the residual.

3. Some defaults: only $4m. A fills, B gets $1m of its $2m, residual gets nothing.

4. Many defaults: $2m. Even A comes up short. B and residual get nothing.

5. Rule: losses hit from the bottom up. Senior is protected by everyone below it.

Worked example

This period A is owed $3m and B is owed $2m. The pool pays (a) $6m, (b) $4m, (c) $2m. Who gets what?

1. (a) $6m: A gets 3, B gets 2, residual gets 1

2. (b) $4m: A gets 3, B gets 1 (short by 1), residual 0

3. (c) $2m: A gets 2 (short by 1), B 0, residual 0

4. So the residual is hit first, then B, and A only in bad cases.

(a) 3/2/1 (b) 3/1/0 (c) 2/0/0

Common mistakes

1. Paying tranches pro rata instead of in order.

2. Forgetting LGD: a defaulted bond still pays its recovery.

3. Assuming correlation always makes every tranche riskier.

Try before checking

Which tranche normally absorbs the first losses?

1. Senior

2. Equity

3. All equally

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_REVIEW_04_Mini_Project_3_CDO_Risk_Part_1_Instructions.txt; FRE6103_NOTES_01_Project_and_recording_notes.md; FRE6103_LECTURE_10_Lecture_03_Corporate_Bonds_and_CDOs.pdf

Lesson 10: Volatility smiles and implied densities

If you back out σ from real option prices, it is not the same for every strike. That pattern tells you what the market really thinks future prices look like.

Start with an everyday idea

Ask the option-pricing recipe a backward question: which volatility would make its price match the market price? Different strikes can give different answers.

Words you need

Implied volatility is inferred from an option price. A smile is the pattern across strikes. A surface adds time to expiry. An implied density describes pricing probabilities, not a guaranteed forecast.

Helpful earlier topics: Black-Scholes and the Greeks

Build the explanation

Implied volatility: the σ you plug into Black-Scholes to match a market price. Solve it with Goal Seek.

Smile / skew: plot implied vol against strike. For stocks, low strikes usually have higher vol (a skew): crashes are feared more than the lognormal model allows.

Volatility surface: implied vol across strike and maturity.

Implied density (Breeden-Litzenberger): the second derivative of call price with respect to strike gives the market's risk-neutral probability density. In practice: a butterfly spread.

European vs Asian: an Asian option pays on the average price, which is less volatile, so it is cheaper. It usually needs simulation.

Formula and units

Implied density: $q(K) = e^{(rT)} \times \partial^2 C/\partial K^2$

Butterfly approx: $q(K) \approx e^{(rT)} \times [C(K-h) - 2 \cdot C(K) + C(K+h)] / h^2$

C(K) = call price at strike K, h = gap between strikes.

Walk through the sketch

1. Black-Scholes assumes one volatility for all strikes: a flat line.

2. Real markets show a skew: low strikes (crash protection) have much higher implied vol.

3. A butterfly (buy 95, sell two 100s, buy 105) pays a little tent. Its cost measures probability near 100.

4. Line up many butterflies and you get the implied density. Compared with the lognormal (dashed), the left tail is fatter.

Worked example

Calls cost 8.00, 5.00 and 2.80 at strikes 95, 100, 105. Estimate the probability density at 100 (ignore discounting). Roughly what is the chance S_T ends between 97.5 and 102.5?

1. Butterfly cost: 8.00 - 2 × 5.00 + 2.80 = 0.80

2. Divide by $h^2 = 25$: density ≈ 0.032 per $1

3. Over a $5 window: 0.032 × 5 = 0.16

4. So about a 16% risk-neutral chance of ending near $100.

Density ≈ 0.032 per $; about 16% chance

Common mistakes

1. Reading a risk-neutral density as a real-world forecast.

2. A negative butterfly cost means arbitrage or bad data, not a negative probability.

3. Using one σ for every strike when the smile is steep.

Try before checking

Implied volatility comes from…

1. Only past stock returns

2. An observed option price and a pricing model

3. A guaranteed future price

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_15_Volatility_Surface.pdf; FRE6103_LECTURE_09_Lecture_9_BSM_and_Implied_Densities.pdf; FRE6103_DATA_01_European_and_Asian_option_benchmarking.xlsx

Lesson 11: Real options and the Merton model

Choices in a business (wait, expand, abandon) are options, so they have value that plain NPV misses. Even a company's stock is an option on its assets.

Start with an everyday idea

A business can wait before building a factory, just as an option holder can wait before exercising. Flexibility can have value when the future is uncertain.

Words you need

A real option is flexibility over a business decision. Merton models a company’s equity like a call on its assets, with debt due at a future date.

Helpful earlier topics: The four derivatives and their payoffs, Credit risk: default, loss and spreads, Black-Scholes and the Greeks

Build the explanation

Real option: the right, not the duty, to take a business action later. Waiting lets you invest only if things go well.

Value it like Class 16: build a tree of the project value, find risk-neutral p, roll back.

If the option value is bigger than investing now (NPV today), wait.

Energy example (VL14): a power plant is a call on the spread between power and fuel prices: run it only when profitable.

Merton model: equity = call option on the firm's assets with strike = debt. Default happens if assets end below the debt. Distance to default counts how many σ the firm is from that line.

Formula and units

Risk-neutral p = [V0·(1 + rf) - V_down] / (V_up - V_down)

Option value = [p·max(V_up - I, 0) + (1-p)·max(V_down - I, 0)] / (1 + rf)

Merton: Equity = V·N(d1) - D·e^(-rT)·N(d2)

DD = [ln(V/D) + (μ - σ²/2)T] / (σ√T), PD = N(-DD)

V = value of project or firm assets, I = investment cost, D = face value of debt.

Walk through the sketch

1. Today the project is worth $105 and costs $100. NPV = $5.

2. Next year it is worth $140 (good) or $80 (bad).

3. If you wait, you invest only in the good case: gain $40. In the bad case you walk away: $0.

4. Rolled back with p = 0.504: the right to wait is worth $19.21. Waiting wins.

5. Same idea for a firm: shareholders get max(Assets - Debt, 0). Equity is a call on the assets.

Worked example

A project is worth $105 today and costs $100. Next year it will be worth $140 or $80; rf = 5%. Invest now, or keep the right to invest next year (cost still $100)?

1. NPV now = 105 - 100 = 5

2. p = (105 × 1.05 - 80) / (140 - 80) = 30.25/60 = 0.504

3. Wait: next year invest only if up: payoff 40 or 0

4. Option value = 0.504 × 40 / 1.05 = 19.21. Bigger than 5, so wait.

5. Merton check: V = 100, D = 80, μ = 6%, σ = 25%, T = 1 → DD = (0.2231 + 0.0288)/0.25 = 1.01, PD = N(-1.01) ≈ 15.7%

Waiting is worth $19.21 vs $5 now. Merton PD ≈ 15.7%

Common mistakes

1. Using the real-world probability in the roll-back.

2. Comparing the option with zero instead of with investing now.

3. In Merton, using the stock's volatility as the asset volatility.

Try before checking

A factory project can wait one year for more information. That flexibility…

1. May have value

2. Must be worthless

3. Guarantees profit

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf; FRE6103_LECTURE_14_VL14_Real_Options_in_Energy.pdf; FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf; Risk syllabus RMN6 (Merton, distance to default)

Paper lab: Move money through time

A promised $121 arrives in two years. The annual rate is 10%. Move it backward one year at a time.



Predict and try

1. Predict: should today’s value be greater or smaller than $121?

2. Compute PV for rates 0%, 5% and 10%. Keep the two-year date fixed.

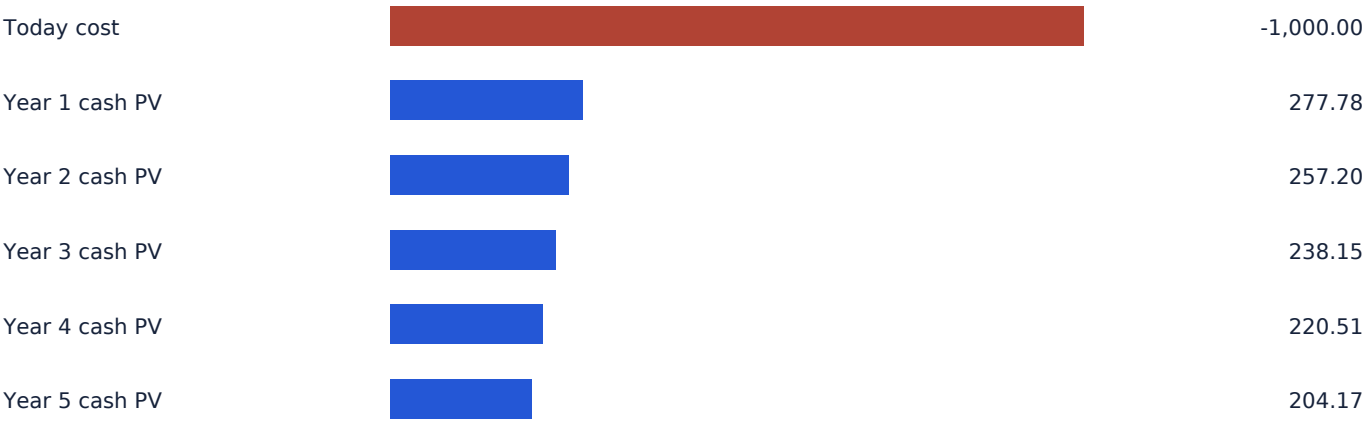
Compare with the explanation

PV = 121/(1+r)^2: $121.00 at 0%, $109.75 at 5%, $100.00 at 10%. A higher discount rate lowers PV. Check: $100 × 1.1 × 1.1 = $121.

Model assumptions: Known future payment; annual compounding; same rate for both years.

Paper lab: Find the rate at which a project breaks even

Pay $1,000 now and receive $300 at each year-end for five years. Discount at 8%.



Predict and try

1. Compute the PV of each future payment and subtract the initial cost.

2. Raise the rate to 20%. What changes? What happens for a one-year cost of $100 and payment of $100 or $50?

Compare with the explanation

At 8%, PV of inflows = $1,197.81 and NPV = +$197.81. At 20%, NPV = -$102.82. The five-year project’s IRR is approximately 15.2382%. The one-year examples have IRRs of 0% and -50%, respectively. Zero NPV means exactly earning the required rate; a negative IRR is a valid losing return.

Model assumptions: Equal end-year payments, one initial outflow, positive later inflows. This cash-flow pattern has a unique IRR greater than -100%; more complex patterns can behave differently.

Paper lab: Change the yield while coupons stay fixed

A bond pays a $50 coupon annually for five years and repays $1,000 with its final coupon.



Predict and try

1. Predict the price relative to face value when yield equals the 5% coupon rate.

2. At 7%, discount all five coupons and the $1,000 principal. Explain why the price falls.

Compare with the explanation

Prices are $1,091.59 at 3%, $1,000.00 at 5%, and $918.00 at 7%. Higher yield discounts the unchanged cash flows more heavily. The last payment is $1,050, not just the $50 coupon.

Model assumptions: Annual coupons, flat yield, no default, valuation immediately after a coupon date.

Paper lab: Build the required return from two pieces

The risk-free rate is 4%; the expected market return is 9%. The market risk premium is 5 percentage points.



Predict and try

1. Calculate required returns for beta 0, 1 and 1.2.

2. Does beta 0 mean the stock has no risk of its own?

Compare with the explanation

E[R] = 4% + beta × (9% - 4%). Returns are 4%, 9% and 10% for beta 0, 1 and 1.2. Beta 0 means zero covariance with the market in this model, not zero total risk. Diversifiable company-specific risk can still exist.

Model assumptions: One-period CAPM with specified inputs. Expected returns are not guaranteed realized returns.

Lesson-check answers

1. Money today vs money later

Answer: $100. Divide $110 by 1.10 to undo one year of growth: $100. Multiplying would move money forward in time.

2. Loans: payments, default and prepayment

Answer: $85. Start with $100, add $5 interest, subtract the $20 payment: $85 remains.

3. Bonds and building a zero curve

Answer: $100. There is one payment: 110 ÷ 1.10 = 100. More payment dates mean more discounted amounts to add.

4. Capital budgeting: NPV and IRR

Answer: $0. PV = 110 ÷ 1.10 = 100. Subtract the $100 cost: NPV = 0.

5. Simulation, correlation and Cholesky

Answer: To represent how assets move together. Independent draws would miss co-movement. Correlation matters for the risk of the combined portfolio.

6. Random walks and stock price paths

Answer: They use different random shocks. The same model can produce many paths because each path receives different random shocks.

7. Portfolio theory and the CAPM

Answer: 13%. Required return = 3% + 2 × 5% = 13%. Beta multiplies the premium, not the risk-free rate.

8. ETFs and futures for end users

Answer: More cash may be needed after adverse moves. Margin is collateral. Adverse daily settlement can require extra cash; losses are not capped at the initial margin.

9. CDO waterfalls and tranches

Answer: Equity. Equity is the first-loss slice. Senior protection can still be exhausted by sufficiently large losses.

10. Volatility smiles and implied densities

Answer: An observed option price and a pricing model. You solve backward from price to volatility. Historical volatility instead measures past variation.

11. Real options and the Merton model

Answer: May have value. Waiting can avoid a bad investment, though delay can also lose revenue. The value depends on assumptions.

Valuation midterm-style practice

7 questions | 100 points | Learn at your own pace.

Original practice, not an official final or a prediction of exam scope. Confirm current course rules. Questions come first; worked solutions and marking guides follow all questions. The suggested time and point weights for mock finals are study choices.

1. Derive a cash-flow value (15 points)
2. Build a Lintner CAPM table (15 points)
3. Correlate and uncorrelate random draws (15 points)
4. Tell uncertainty from simulation noise (10 points)
5. Critique a correlation claim (15 points)
6. Write a process with reversion and jumps (15 points)
7. Explain capital and performance (15 points)

Question 1: Derive a cash-flow value

15 points | 2025A Q1 · MBE/GVE and cash-flow series

Payments starting one period from now are 0, 2, 6, 12, 20, 30, … forever. Each payment increases by 2 more than the previous increase. Derive the present value V at a constant per-period discount rate r > 0. Show the shift-and-subtract argument, not just a memorized answer. Then value the stream at r = 10%.

Show your setup, units, calculation and interpretation.

Question 2: Build a Lintner CAPM table

15 points | 2025A Q2 · Cash-flow covariance and valuation

Two securities have expected next-period cash flows 40 and 60. Their covariance matrix is [[64,16],[16,144]]. The risk-free rate is zero. Asset 1 is worth 36. Using Vi = E[Xi] - A Cov(Xi, XM) with XM = X1 + X2, find A, asset 2’s value, total market value, value weights, expected returns and return standard deviations.

Show your setup, units, calculation and interpretation.

Question 3: Correlate and uncorrelate random draws

15 points | 2025A Q3 · Cholesky and matrix orientation

Find a lower-triangular Cholesky factor of Σ = [[25,15],[15,45]]. If each row of an n×2 matrix U contains independent standard-normal draws, how do you obtain correlated zero-mean rows W? How do you undo the transformation?

Show your setup, units, calculation and interpretation.

Question 4: Tell uncertainty from simulation noise

10 points | 2025A Q4 · Standard error and sample-size scaling

A simulation has n = 400 and maxerr = 4.00. Use the supplied exam key’s convention maxerr = 2 × SE. Find SE and sample standard deviation s. How many independent draws are needed for maxerr at most 1.50, assuming s stays the same?

Show your setup, units, calculation and interpretation.

Question 5: Critique a correlation claim

15 points | 2025A Q5 · Reasoning from data

An analyst shows two asset-price series rising through a year and reports a level correlation of +0.65. They conclude the assets respond to the same economic forces and provide little diversification. Write a short critique and describe a better analysis. You are not given the return data.

Show your setup, units, calculation and interpretation.

Question 6: Write a process with reversion and jumps

15 points | 2025A Q6 · Translate a story into a model

Write a continuous-time price model with mean reversion toward μ at speed κ, diffusion volatility σ and upward jumps arriving at intensity π. At each jump, size J is exponential with rate λ with probability 40%, or a fixed $3 with probability 60%. Give an Euler simulation step and explain all random draws.

Show your setup, units, calculation and interpretation.

Question 7: Explain capital and performance

15 points | 2025A Q7 · Short written explanation

Explain why capital differs from cash, why a bank needs a loss-absorbing cushion, and why comparing profit alone can mislead when activities take different risks. State how the supplied course key describes risk-adjusted performance, and distinguish that from common return-on-economic-capital terminology.

Show your setup, units, calculation and interpretation.

Solution 1: Derive a cash-flow value

First, the words

A perpetuity is a payment stream with no last payment. GVE relates today's value to the next payment and the value remaining afterward. Here r is a positive rate per period.

A smaller problem first

For $2 every period forever, starting next period, value is 2/r. At 10%, that is $20. Why? The first payment plus the remaining value, discounted once, must equal today's value: V = (2 + V)/(1+r).

Hint

Write V for the original stream and Vnext for the remaining 2, 6, 12, 20, … stream. The difference Vnext - V is 2, 4, 6, … . First value that simpler increasing stream.

Work through it

1. Let L be the value of 2,4,6,… . After one period its remaining stream is 4,6,8,…, worth Lnext. The difference Lnext - L is a flat $2 perpetuity, worth 2/r.

2. Apply GVE to L: $(1+r)L = 2 + Lnext = 2 + L + 2/r$. Therefore $L = 2(1+r)/r^2$.

3. Now let V be the value of 0,2,6,12,… . Its first payment is zero, so $(1+r)V = Vnext$. The shifted stream minus the original stream is 2,4,6,…, so Vnext - V = L.

4. Subtract V from the GVE: $rV = L$. Hence $V = 2(1+r)/r^3$.

5. At r = 0.10, $V = 2 \times 1.10 / 0.001 = \$2{,}200$. Timing matters: the first payment at t = 1 is zero. The infinite series converges for r > 0.

Mark your reasoning

1. 4: timeline and GVE for the original stream

2. 5: values the first-difference stream

3. 4: formula and $2,200

4. 2: r > 0 and first-payment timing

Shimko, Valuation Principles Chapters 1-2: MBE/GVE and growing payment patterns. This follows the supplied triangular-series question with scaled payments.

My mark: ____ / 15. My next correction: _____

Solution 2: Build a Lintner CAPM table

First, the words

Mean is the expected future cash flow. Variance is a measure of spread; covariance measures how two cash flows vary together. A is the course model's price per unit of covariance risk.

A smaller problem first

With no risk-free interest, value = expected cash flow - risk charge. If the mean is $50 and the charge is $6, value is $44. This question uses the course's cash-flow CAPM equation, not just a beta/return shortcut.

Hint

Cov(X1,XM) is the sum of row 1, not just asset 1's variance. Return standard deviation is cash-flow standard deviation divided by today's value.

Work through it

1. Row sums: Cov(X1,XM) = 64 + 16 = 80; Cov(X2,XM) = 16 + 144 = 160.

2. Use asset 1: 36 = 40 - 80A, so A = 4/80 = 0.05. Asset 2 value = 60 - 0.05 × 160 = 52.

3. Market value M = 36 + 52 = 88. Value weights are 36/88 = 40.91% and 52/88 = 59.09%.

4. Expected returns: 40/36 - 1 = 11.11%; 60/52 - 1 = 15.38%; market = 100/88 - 1 = 13.64%.

5. Cash-flow standard deviations: √64 = 8 and √144 = 12. Divide by values: 8/36 = 22.22%, 12/52 = 23.08%.

6. Market cash-flow variance is the sum of all four matrix entries: 240. Its return standard deviation is √240 / 88 = 17.60%.

Mark your reasoning

1. 4: row sums and A = 0.05

2. 3: values and weights

3. 4: expected returns

4. 4: standard deviations including market covariance

Valuation Lecture 6: MPT and the CAPM; the supplied Sharpe/Lintner spreadsheet and the 2025A key. This is a course-specific one-period setup.

My mark: ____ / 15. My next correction: _____

Solution 3: Correlate and uncorrelate random draws

First, the words

A matrix is a rectangular table of numbers. Transpose (T) swaps rows and columns. An inverse reverses a transformation. Cholesky finds a lower-triangular matrix C with $\Sigma = CC^T$.

A smaller problem first

For a lower-triangular 2×2 matrix C = [[a,0],[b,c]], multiplying C by C^T gives $[[a^2,ab],[ab,b^2+c^2]]$. Match these entries to the covariance matrix one at a time.

Hint

Start with $a^2 = 25$. Then $ab = 15$. Finish with $b^2 + c^2 = 45$. Pay attention to whether observations are stored in rows or columns.

Work through it

1. $a = \sqrt{25} = 5$. $b = 15/5 = 3$. $c = \sqrt{(45 - 3^2)} = \sqrt{36} = 6$.

2. $C = [[5,0],[3,6]]$. Check $CC^T = [[25,15],[15,9+36]] = \Sigma$.

3. For observations in rows, $W = UC^T$. Row transformation gives the required covariance.

4. Undo it with $U = W(C^T)^{-1}$. If W has nonzero means, subtract those means first.

5. A valid covariance matrix must be positive semidefinite. This example is positive definite; the diagonal square roots are positive and the inverse exists.

Mark your reasoning

1. 7: factor entries

2. 3: multiplication check

3. 3: forward transformation

4. 2: inverse and orientation

Shimko simulation excerpt: multivariate simulation and Cholesky; Valuation Lecture 4 and Cholesky spreadsheet.

My mark: ____ / 15. My next correction: _____

Solution 4: Tell uncertainty from simulation noise

First, the words

Standard deviation (s) describes spread in individual simulated outcomes. Standard error (SE) describes uncertainty in their average. More draws reduce SE, not the underlying risk of each outcome.

A smaller problem first

SE = s/√n. If s = 20 and n = 100, then SE = 2. Four times as many independent draws cut the standard error in half.

Hint

First halve maxerr. Then multiply SE by √n. Sample size grows with the square of the ratio of old to new error.

Work through it

1. SE = 4/2 = 2.00.

2. s = SE × √400 = 2 × 20 = 40.00.

3. nnew = 400 × (4/1.5)² = 2,844.44…. Round upward to 2,845 draws.

4. The factor 2 is the convention used here, approximately a 95% normal interval. Dependence or poor simulation assumptions can invalidate this simple error estimate.

Mark your reasoning

1. 3: SE = 2

2. 3: s = 40

3. 3: 2,845 draws, rounded up

4. 1: independent-draw and unchanged-spread assumptions

Valuation Lecture 4: Simulation; supplied 2025A Q4/key. Its numbers show the maxerr = 2 SE convention.

My mark: ____ / 10. My next correction: _____

Solution 5: Critique a correlation claim

First, the words

A price level is a price at a date. A return measures the change relative to the previous price. A shared trend can create a strong level correlation without synchronized short-term changes.

A smaller problem first

Two people can both walk uphill while one pauses whenever the other speeds up. Their heights can rise together even when their step-by-step changes do not.

Hint

Separate what the chart shows, what it does not establish, and what you would calculate next. Do not invent a return correlation.

Work through it

1. The correlation is between levels, which can be persistent or trending. Shared trends can drive the statistic without establishing a causal relationship.

2. Calculate aligned period-by-period returns or changes, choose an economically relevant frequency, and inspect their scatter plot and correlation.

3. Check missing dates, outliers, sample length, changing relationships and serial dependence. Returns are not automatically independent or identically distributed.

4. Diversification depends on the joint behavior of portfolio returns, exposures and the horizon. A single level-correlation number is insufficient.

5. We cannot infer the sign or magnitude of return correlation without the actual data. State that limitation rather than claiming it must be negative.

Mark your reasoning

1. 5: identifies level/trend problem

2. 4: proposes returns/changes analysis

3. 3: limits causal/diversification claim

4. 3: acknowledges missing data and model checks

Supplied 2025A Q5/key; Valuation stochastic-process and simulation materials. The particular sign discussed in the source key is not a universal rule.

My mark: ____ / 15. My next correction: _____

Solution 6: Write a process with reversion and jumps

First, the words

Mean reversion pulls the price toward a long-run level μ. κ controls the speed. Brownian shocks model small random moves. A Poisson process N counts occasional jump arrivals with intensity π per unit time.

A smaller problem first

If the current price is 80 and the long-run level is 100, κ(100 - 80) is positive: the drift points upward. If the price is 120, that drift points downward.

Hint

Separate whether a jump occurs from how large that jump is. Brownian increments scale with √Δt; drift scales with Δt.

Work through it

1. A compound-Poisson version is dS = κ(μ - S)dt + σdW + J dN, with independently sampled marks J at each jump and N a Poisson counting process of intensity π.

2. For each jump, draw B from Bernoulli(0.4). If B = 1, use J = -ln(U)/λ with U uniform on (0,1); otherwise use J = 3. λ must be positive.

3. For one simulation step, draw Z standard normal and K Poisson(πΔt), independently. Draw K jump sizes J1,...,JK from the mixture.

4. Euler step: Snext = S + κ(μ - S)Δt + σ√Δt Z + Σj Jj. If K = 0 the sum is zero.

5. For very small πΔt, a Bernoulli jump indicator with probability approximately πΔt is a small-step approximation. It misses multiple arrivals; do not use it as an exact large-step rule.

6. This model has upward jumps and an additive diffusion, which can permit negative prices. Mean reversion is the drift component; jumps shift the overall stationary mean unless compensated. State assumptions.

Mark your reasoning

1. 4: mean-reverting drift and diffusion

2. 3: arrival process

3. 4: mixture and exponential sampling

4. 4: time scaling and separate independent draws

Valuation Lecture 5: Stochastic Processes; Hull 8e Chapter 13 for Wiener-process foundations. The source key has compressed/inconsistent symbols; this practice uses explicit μ for mean and π for jump intensity.

My mark: ____ / 15. My next correction: _____

Solution 7: Explain capital and performance

First, the words

Cash is a liquid asset used to pay bills. Capital is a loss-absorbing funding cushion; economic capital is a model-based amount set aside against risk. They are not interchangeable.

A smaller problem first

A bank can hold many valuable long-term assets but little cash today. It can also have plenty of cash financed by debt and very little equity. Liquidity and loss absorption are different questions.

Hint

Write one sentence each for definition, banking purpose, comparable risk-adjusted performance and metric conventions.

Work through it

1. Cash concerns immediate payment ability. Capital absorbs losses and supports risk-taking; economic capital estimates a cushion for adverse losses under a chosen model and confidence level. It is not a guaranteed maximum-loss bound.

2. Banks need loss-absorbing resources because credit, market and operational losses can otherwise threaten creditors and continued operation.

3. Two desks with the same profit may use very different amounts of risk capital. Compare performance after accounting for funding and risk, using consistent definitions and time horizons.

4. The supplied 2025A key names the Sharpe ratio or dollarized Sharpe ratio. Learn that course wording: the usual Sharpe ratio measures excess return per unit of return standard deviation.

5. Common industry usage also includes RAROC/return on economic capital, which uses a capital denominator with a specified income definition. These measures are related ideas, not identical formulas. State the convention requested by the instructor.

Mark your reasoning

1. 4: cash versus capital

2. 4: loss absorption and bank risk

3. 4: risk-adjusted comparison

4. 3: accurate course wording and distinction between metrics

Supplied 2025A Q7/key; Risk syllabus and Valuation capital/performance discussion. Ask the instructor to confirm metric terminology for the current assessment.

My mark: ____ / 15. My next correction: _____

Valuation mock final

8 questions | 100 points | 120 minutes suggested practice time.

Original practice, not an official final or a prediction of exam scope. Confirm current course rules. Questions come first; worked solutions and marking guides follow all questions. The suggested time and point weights for mock finals are study choices.

1. Derive an increasing cash-flow value (12 points)
2. Bootstrap and check interest-rate sensitivity (12 points)
3. Price an option using a cash-flow risk adjustment (12 points)
4. Audit a multivariate simulation (12 points)
5. Read a probability clue from option prices (12 points)
6. Compare committing now with waiting (14 points)
7. Connect financing assumptions to capital budgeting (14 points)
8. Trace losses and choose a risk response (12 points)

Question 1: Derive an increasing cash-flow value

12 points | Revised syllabus week 1; historical midterm derivation style.

Cash flows at years 1, 2, 3, ... are $10, $20, $30, ... forever. Derive V using a shift and GVE, then evaluate it at r = 10% annually. Explain why using the same rate for uncertain future cash flows needs an additional assumption.

Show your setup, units, calculation and interpretation.

Question 2: Bootstrap and check interest-rate sensitivity

12 points | Revised syllabus weeks 2-3: bonds, zero curve and interest-rate risk.

The one-year annual spot rate is 4%. A two-year annual-coupon bond with face $100 and coupon 5% costs $100. Derive D2 and the two-year annually compounded spot rate. Separately a bond portfolio is worth $20,000 with modified duration 4.2. Estimate its value change if its yield rises by 0.50 percentage points. State what you are ignoring.

Show your setup, units, calculation and interpretation.

Question 3: Price an option using a cash-flow risk adjustment

12 points | Revised syllabus weeks 6 and 9: Lintner CAPM and option valuation.

There are two equally likely physical states. Market cash flow XM is 80 or 120; asset cash flow X is 70 or 130 in the same states. A call pays Y = max(X - 100, 0). Use V(Y) = [E(Y) - A Cov(Y,XM)]/(1 + rf), with A = 0.01 and rf = 5%. Find the call’s expected cash flow, covariance, risk-adjusted value and value if the covariance charge were wrongly omitted. Explain the difference.

Show your setup, units, calculation and interpretation.

Question 4: Audit a multivariate simulation

12 points | Revised syllabus weeks 4-5; historical midterm computational style.

Find a lower-triangular C for covariance matrix [[9,6],[6,20]]. State the transformation from independent standard-normal rows U to correlated rows W and its inverse. A separate simulation gives payoff standard deviation s = 24 from n = 144 independent draws. With maxerr = 2SE, find its maxerr and the number of draws needed for maxerr no greater than 1, keeping s unchanged.

Show your setup, units, calculation and interpretation.

Question 5: Read a probability clue from option prices

12 points | Revised syllabus week 10: implied probability distributions.

Same-expiry European calls on a non-dividend stock have prices C(90) = 16, C(100) = 9, C(110) = 4. The expiry discount factor is D = 0.95. Find the cost of the 90/100/110 butterfly, its maximum terminal payoff, and a centered finite-difference estimate of the risk-neutral density near strike 100 using $fQ(K) \approx [C(K-h) - 2C(K) + C(K+h)]/(D\,h^2)$, h = 10. Interpret fQ × h carefully. What fails if C(100) were 11 instead?

Show your setup, units, calculation and interpretation.

Question 6: Compare committing now with waiting

14 points | Revised syllabus week 11: contracts and real options.

Next year a project asset will be worth 160 or 80. Treat these as market-consistent asset values with risk-neutral probabilities 0.5 each. The annual risk-free growth factor is 1.05. You can either pay 100 now to own the asset next year, or hold a free right to pay 100 next year after seeing its value. Find immediate-investment NPV, the value of waiting and the incremental flexibility value. Explain why using this method for a nontraded business needs care.

Show your setup, units, calculation and interpretation.

Question 7: Connect financing assumptions to capital budgeting

14 points | Revised syllabus weeks 11-12: capital budgeting and capital structure.

A company has market equity value 600 and debt 400. Its cost of equity is 12%, pre-tax cost of debt is 6%, and corporate tax rate is 25%. Under a stable target financing mix and comparable project risk, calculate after-tax WACC. A project costs 100 now and pays one unlevered after-tax cash flow of 109 next year. Find NPV and IRR, and explain the decision at this rate. Why can you not keep the same component costs while increasing debt indefinitely to make value rise?

Show your setup, units, calculation and interpretation.

Question 8: Trace losses and choose a risk response

12 points | Revised syllabus: corporate bonds/CDOs and later corporate risk management.

A $100 debt pool has equity 5, mezzanine 10 and senior 85. Allocate a realized $12 pool loss. Separately a company’s annual loss distribution is 0 with probability 90%, 10 with probability 8%, and 40 with probability 2% (all in $m). Find expected loss, 95% VaR as the smallest loss with cumulative probability at least 95%, and ES as the average of the worst 5%, splitting the mass at the VaR threshold as needed. Give one reason management should examine stress scenarios as well.

Show your setup, units, calculation and interpretation.

Solution 1: Derive an increasing cash-flow value

First, the words

GVE links today's value to the next cash flow and remaining value. A perpetuity continues indefinitely; the discount rate here is positive and per year.

A smaller problem first

A flat $10 perpetuity beginning next year is worth 10/r. Use a shifted stream to handle payments that grow by a fixed dollar amount.

Hint

After the first year, the remaining stream is the original stream plus $10 each year.

Work through it

1. Let Vnext value the stream 20,30,40,... one year later. Relative to 10,20,30,... it adds a $10 perpetuity, so Vnext = V + 10/r.

2. GVE: (1 + r)V = 10 + Vnext = 10 + V + 10/r.

3. Therefore rV = 10 + 10/r and V = 10(1 + r)/r^2.

4. At r = 0.10, V = 10 × 1.1/0.01 = $1,100. The first payment is at year 1 and r > 0 ensures convergence.

5. For uncertain cash flows, their risk and timing must be priced consistently. A risk-free rate applied to raw expected risky cash flows generally omits a risk adjustment.

Mark your reasoning

1. 4: shifted stream

2. 3: GVE and formula

3. 3: value, timing and domain

4. 2: risk limitation

Shimko Valuation Principles Chapters 1-2 and Lecture 1.

My mark: ____ / 12. My next correction: _____

Solution 2: Bootstrap and check interest-rate sensitivity

First, the words

Bootstrapping finds each discount factor from traded cash flows. Duration is a first-order approximation, not an exact prediction for large yield moves.

A smaller problem first

A one-year zero paying $100 at 4% annually costs 100/1.04. That gives D1 = 1/1.04.

Hint

Price = 5D1 + 105D2. For duration, convert 0.50 percentage points to 0.005.

Work through it

1. D1 = 1/1.04 = 0.96153846.

2. 100 = 5D1 + 105D2, so D2 = (100 - 5/1.04)/105 = 0.90659341.

3. Two-year spot rate z2 = D2^(-1/2) - 1 = 5.02525% annually.

4. Approximate change = -modified duration × value × yield change = -4.2 × 20,000 × 0.005 = -$420; estimated new value $19,580.

5. The approximation ignores convexity and changes in cash flows; a single yield shift also does not capture arbitrary curve twists or spread changes.

Mark your reasoning

1. 4: bootstrapping setup

2. 3: D2 and spot rate

3. 3: duration calculation

4. 2: limitations

Valuation bond/interest-rate lecture; zero-coupon term-structure mini-project topics.

My mark: ____ / 12. My next correction: _____

Solution 3: Price an option using a cash-flow risk adjustment

First, the words

An option’s cash flow is nonlinear. Compute its own covariance with the market before applying the supplied cash-flow pricing model.

A smaller problem first

A call paying max(X - 100, 0) on asset cash flows 70 and 130 pays 0 and 30, not -30 and 30.

Hint

Covariance is the average of the products of deviations from each variable’s mean.

Work through it

1. Y is 0 in the low state and 30 in the high state. E(Y) = 15; E(XM) = 100.

2. Cov(Y,XM) = 0.5(-15)(-20) + 0.5(15)(20) = 300.

3. Risk charge = 0.01 × 300 = 3. Call value = (15 - 3)/1.05 = $11.4286.

4. Ignoring risk gives 15/1.05 = $14.2857, an overstatement of $2.8571 in this stipulated model.

5. The option pays in the high-market state, so the model charges for positive market covariance. This is a given one-period model, not an assertion that physical probabilities equal risk-neutral probabilities.

Mark your reasoning

1. 2: payoffs and mean

2. 4: covariance

3. 3: adjusted price

4. 3: comparison and interpretation

Valuation Lecture 6 and revised syllabus week 9; follow your instructor’s notation and calibration.

My mark: ____ / 12. My next correction: _____

Solution 4: Audit a multivariate simulation

First, the words

Cholesky creates correlated draws. Standard error measures uncertainty in an estimated mean; it is different from the standard deviation of individual outcomes.

A smaller problem first

For $C = [[a,0],[b,c]]$, $CC^T = [[a^2,ab],[ab,b^2+c^2]]$. With observations in rows, transform independent draws using C^T.

Hint

Match a^2, ab and $b^2 + c^2$. Then use SE = s/sqrt(n).

Work through it

1. a = 3, b = 2, c = 4. Thus $C = [[3,0],[2,4]]$. Check $CC^T = [[9,6],[6,20]]$.

2. For row observations $W = UC^T$; recover $U = W(C^T)^{-1}$, after subtracting means if needed.

3. SE = 24/sqrt(144) = 2. Therefore maxerr = 4.

4. Need $2 \times 24/\sqrt{n} \leq 1$, so $n \geq 48^2 = 2{,}304$.

5. Independence and unchanged outcome spread are assumptions. More draws reduce sampling error, not model bias or the underlying payoff risk.

Mark your reasoning

1. 4: Cholesky and check

2. 3: row transformation and inverse

3. 3: error and sample size

4. 2: limitations

Valuation simulation lecture and Cholesky spreadsheet; historical maxerr convention is specified explicitly.

My mark: ____ / 12. My next correction: _____

Solution 5: Read a probability clue from option prices

First, the words

A butterfly combines calls at three equally spaced strikes. Call-price curvature contains information about risk-neutral probabilities, not directly real-world probabilities.

A smaller problem first

A long butterfly buys the low-strike call, sells two middle-strike calls and buys the high-strike call. Its payoff is never negative.

Hint

Positive curvature is required by no arbitrage. A density has units of probability per dollar.

Work through it

1. Butterfly cost = 16 - 2 × 9 + 4 = $2.

2. The payoff rises from 0 at S = 90 to $10 at S = 100 and falls to 0 at S = 110; outside that range it is zero.

3. Density estimate = 2/(0.95 × 100) = 0.0210526 per dollar.

4. Multiplying by a $10 interval gives roughly 0.2105, or 21.05%, only if a local constant-density approximation is reasonable. This coarse estimate is not an exact interval probability.

5. If the middle call were 11, butterfly cost would be -$2 while terminal payoff remains nonnegative. That violates convexity and allows an arbitrage under frictionless trading.

6. These are pricing-measure probabilities. Risk premia and the finite strike spacing prevent reading them as a precise real-world forecast.

Mark your reasoning

1. 3: cost and payoff

2. 3: density calculation

3. 3: careful interpretation

4. 3: convexity violation

Valuation implied-density and volatility-surface lectures; the formula is supplied to make the exercise self-contained.

My mark: ____ / 12. My next correction: _____

Solution 6: Compare committing now with waiting

First, the words

An option to invest lets you reject a bad outcome. A risk-neutral probability is given here; do not substitute physical forecasts without changing the model.

A smaller problem first

If next year’s project asset is worth 80 and investment costs 100, an optional investment pays 0 because you can decline.

Hint

Discount the asset value for the commit-now case; discount max(asset value - investment cost, 0) for the waiting case.

Work through it

1. Asset PV = (0.5 × 160 + 0.5 × 80)/1.05 = $114.2857.

2. Immediate-investment NPV = 114.2857 - 100 = $14.2857.

3. Waiting payoffs are $60 in the high state and $0 in the low state.

4. Waiting value = (0.5 × 60 + 0.5 × 0)/1.05 = $28.5714.

5. Incremental value relative to committing now = 28.5714 - 14.2857 = $14.2857. It includes the ability to avoid bad investment and defer the expenditure.

6. For a nontraded project, replication may be unavailable. Justify the pricing probabilities/risk adjustment, possible competition and any cost of waiting. The supplied assumptions make this toy model calculable.

Mark your reasoning

1. 3: asset PV and immediate NPV

2. 4: contingent payoffs and option value

3. 3: incremental comparison

4. 4: assumptions and nontraded limitation

Valuation real-options-in-energy lecture; Welch capital budgeting and option intuition.

My mark: ____ / 14. My next correction: _____

Solution 7: Connect financing assumptions to capital budgeting

First, the words

WACC weights financing costs by market values. The tax adjustment here assumes deductible interest and a usable corporate tax shield.

A smaller problem first

If 60% of funding is equity and 40% is debt, multiply each financing cost by its weight; do not average them equally.

Hint

WACC = wE ke + wD kd(1 - tax). One-period IRR solves cost = future cash/(1 + IRR).

Work through it

1. Equity and debt weights are 0.6 and 0.4. After-tax debt cost is 6% × 0.75 = 4.5%.

2. WACC = 0.6 × 12% + 0.4 × 4.5% = 9%.

3. NPV = -100 + 109/1.09 = 0. IRR = 109/100 - 1 = 9%.

4. The project exactly earns the assumed required return: it is financially break-even at these inputs, not a strictly positive-NPV opportunity.

5. Higher leverage changes equity/debt risk and their required returns. Distress costs, tax-shield limits and financing constraints can alter value.

6. Use cash flows and discount rates consistently. Do not subtract interest again from the unlevered cash flow when discounting at WACC.

Mark your reasoning

1. 4: weights and WACC

2. 3: NPV

3. 2: IRR

4. 2: break-even decision

5. 3: leverage and consistency

Welch Corporate Finance: capital budgeting, financing and capital structure. This is a syllabus extension, not a historical midterm replica.

My mark: ____ / 14. My next correction: _____

Solution 8: Trace losses and choose a risk response

First, the words

A tranche absorbs losses according to seniority. A risk measure summarizes a distribution; it does not remove exposure.

A smaller problem first

If the equity tranche absorbs the first $5 of loss, a $3 pool loss never reaches mezzanine or senior investors.

Hint

The worst 5% includes all of the 2% at loss 40 and 3 percentage points from the loss-10 outcomes.

Work through it

1. Equity loses 5, mezzanine loses 7, senior loses 0. Remaining tranche values under this simple loss-only allocation are 0, 3 and 85.

2. Expected loss = 0.90 × 0 + 0.08 × 10 + 0.02 × 40 = $1.6m.

3. Cumulative probability is 90% at 0 and 98% at 10, so 95% VaR = $10m.

4. Worst-5% average = (0.02 × 40 + 0.03 × 10)/0.05 = $22m.

5. Stress testing can examine losses beyond an estimated distribution, correlation breakdowns, illiquidity and funding pressures. It complements a quantified limit rather than proving the model correct.

Mark your reasoning

1. 3: waterfall

2. 2: expected loss

3. 2: VaR

4. 3: ES

5. 2: stress rationale

Valuation corporate bonds/CDOs lecture; later corporate-risk-management syllabus topic. Risk course assessment remains report/presentation based.

My mark: ____ / 12. My next correction: _____